



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date: 24/09/2026

Laboratory Manual

Academic Year	:	2026-27	Semester	:	3rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical – 7

PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

Problem Statement:- Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages.

Objective:- To develop a PHP-based server-side form processing system that validates and sanitizes user inputs, securely stores submitted registration/contact data in CSV or JSON format, and displays appropriate success or error messages.

Github Repository Link:- https://github.com/Nishtha-cse/StudentHubPortal-D26CS129

index.php :

```php
<?php
session_start();

$errors  = $_SESSION['errors']   ?? [];
$oldData = $_SESSION['old_data'] ?? [];
$success = $_SESSION['success']  ?? '';

unset($_SESSION['errors'], $_SESSION['old_data'], $_SESSION['success']);

function old(array $data, string $key): string {
    return htmlspecialchars($data[$key] ?? '', ENT_QUOTES, 'UTF-8');
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Registration</title>

    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link
href="https://fonts.googleapis.com/css2?family=DM+Sans:wght@400;500;600;700&di
splay=swap" rel="stylesheet">
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <main class="form-card">
        <h1>Create your registration</h1>
        <p class="intro">Fill in your details below. All fields are
required.</p>

        <?php if ($success): ?>
            <div class="alert success">
                <?= htmlspecialchars($success) ?>
            </div>
        <?php endif; ?>

        <?php if ($errors): ?>
            <div class="alert error" role="alert">
                <strong>⚠ Please correct the following errors:</strong>
                <ul>
                    <?php foreach ($errors as $error): ?>
                        <li><?= htmlspecialchars($error) ?></li>
                    <?php endforeach; ?>
                </ul>
```

```
            </div>
        <?php endif; ?>

        <!-- novalidate lets the server-side errors above appear instead of
browser popups -->
        <form action="process.php" method="POST" novalidate>
            <div class="form-grid">

                <div class="field">
                    <label for="name">Name</label>
                    <input type="text" id="name" name="name"
autocomplete="name"
                           value="<?= old($oldData, 'name') ?>">
                </div>

                <div class="field">
                    <label for="email">Email</label>
                    <input type="email" id="email" name="email"
autocomplete="email"
                           value="<?= old($oldData, 'email') ?>">
                </div>

                <div class="field">
                    <label for="mobile">Mobile</label>
                    <input type="text" id="mobile" name="mobile"
inputmode="numeric"
                           maxlength="10" autocomplete="tel-national"
                           value="<?= old($oldData, 'mobile') ?>">
                </div>

                <div class="field">
                    <label for="pincode">Pincode</label>
                    <input type="text" id="pincode" name="pincode"
inputmode="numeric"
                           maxlength="6" autocomplete="postal-code"
                           value="<?= old($oldData, 'pincode') ?>">
                </div>

                <fieldset class="field full">
                    <legend>Gender</legend>
                    <div class="radio-row">
                        <input type="radio" id="male" name="gender"
value="male"
                               <?= ($oldData['gender'] ?? '') === 'male' ?
'checked' : '' ?>>
                        <label for="male">Male</label>
```

```
                        <input type="radio" id="female" name="gender"
value="female"
                               <?= ($oldData['gender'] ?? '') === 'female' ?
'checked' : '' ?>>
                        <label for="female">Female</label>
                    </div>
                </fieldset>

                <div class="field full">
                    <label for="address">Address</label>
                    <input type="text" id="address" name="address"
autocomplete="street-address"
                           value="<?= old($oldData, 'address') ?>">
                </div>

                <div class="field full">
                    <label for="message">Message</label>
                    <textarea id="message" name="message"><?= old($oldData,
'message') ?></textarea>
                </div>

            </div>

            <div class="actions">
                <button type="submit">Submit</button>
            </div>
        </form>
    </main>
</body>
</html>
```

Process.php :

```
<?php
session_start();

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    header('Location: index.php');
    exit();
}

$storage_mode = 'json';   // 'json' or 'csv'

/* ---------- Read input ---------- */
$name    = trim($_POST['name']    ?? '');
$email   = trim($_POST['email']   ?? '');
$mobile  = trim($_POST['mobile']  ?? '');
$pincode = trim($_POST['pincode'] ?? '');
```

```php
$gender  = trim($_POST['gender']  ?? '');
$address = trim($_POST['address'] ?? '');
$message = trim($_POST['message'] ?? '');

// Kept so the form can be refilled if validation fails
$oldData = compact('name', 'email', 'mobile', 'pincode', 'gender', 'address',
'message');

/* ---------- Validate ---------- */
$errors = [];

if ($name === '') {
    $errors[] = 'Name is required.';
} elseif (!preg_match('/^[a-zA-Z ]{2,50}$/', $name)) {
    $errors[] = 'Name must contain only letters and spaces.';
}

if ($email === '') {
    $errors[] = 'Email address is required.';
} elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
    $errors[] = 'Enter a valid email address.';
}

if ($mobile === '') {
    $errors[] = 'Mobile number is required.';
} elseif (!preg_match('/^[0-9]{10}$/', $mobile)) {
    $errors[] = 'Mobile number must contain exactly 10 digits.';
}

if ($pincode === '') {
    $errors[] = 'Pincode is required.';
} elseif (!preg_match('/^[0-9]{6}$/', $pincode)) {
    $errors[] = 'Pincode must contain exactly 6 digits.';
}

if (!in_array($gender, ['male', 'female'], true)) {
    $errors[] = 'Please select a valid gender.';
}

if ($address === '') {
    $errors[] = 'Address is required.';
}

if ($message === '') {
    $errors[] = 'Message is required.';
}

if ($errors) {
```

```
    $_SESSION['errors']   = $errors;
    $_SESSION['old_data'] = $oldData;
    header('Location: index.php');
    exit();
}

/* ---------- Sanitize ---------- */
$entry = [
    'timestamp' => date('Y-m-d H:i:s'),
    'name'      => htmlspecialchars($name,    ENT_QUOTES, 'UTF-8'),
    'email'     => htmlspecialchars($email,   ENT_QUOTES, 'UTF-8'),
    'mobile'    => htmlspecialchars($mobile,  ENT_QUOTES, 'UTF-8'),
    'pincode'   => htmlspecialchars($pincode, ENT_QUOTES, 'UTF-8'),
    'gender'    => htmlspecialchars($gender,  ENT_QUOTES, 'UTF-8'),
    'address'   => htmlspecialchars($address, ENT_QUOTES, 'UTF-8'),
    'message'   => htmlspecialchars($message, ENT_QUOTES, 'UTF-8'),
];

/* ---------- Store ---------- */
function saveToCSV(string $file, array $entry): bool
{
    $isNew = !file_exists($file);
    $handle = fopen($file, 'a');
    if (!$handle || !flock($handle, LOCK_EX)) {
        return false;
    }
    if ($isNew) {
        fputcsv($handle, array_keys($entry));
    }
    fputcsv($handle, array_values($entry));
    flock($handle, LOCK_UN);
    fclose($handle);
    return true;
}

function saveToJSON(string $file, array $entry): bool
{
    $records = file_exists($file) ? json_decode(file_get_contents($file),
true) : [];
    if (!is_array($records)) {
        $records = [];
    }
    $records[] = $entry;
    return file_put_contents($file, json_encode($records, JSON_PRETTY_PRINT),
LOCK_EX) !== false;
}

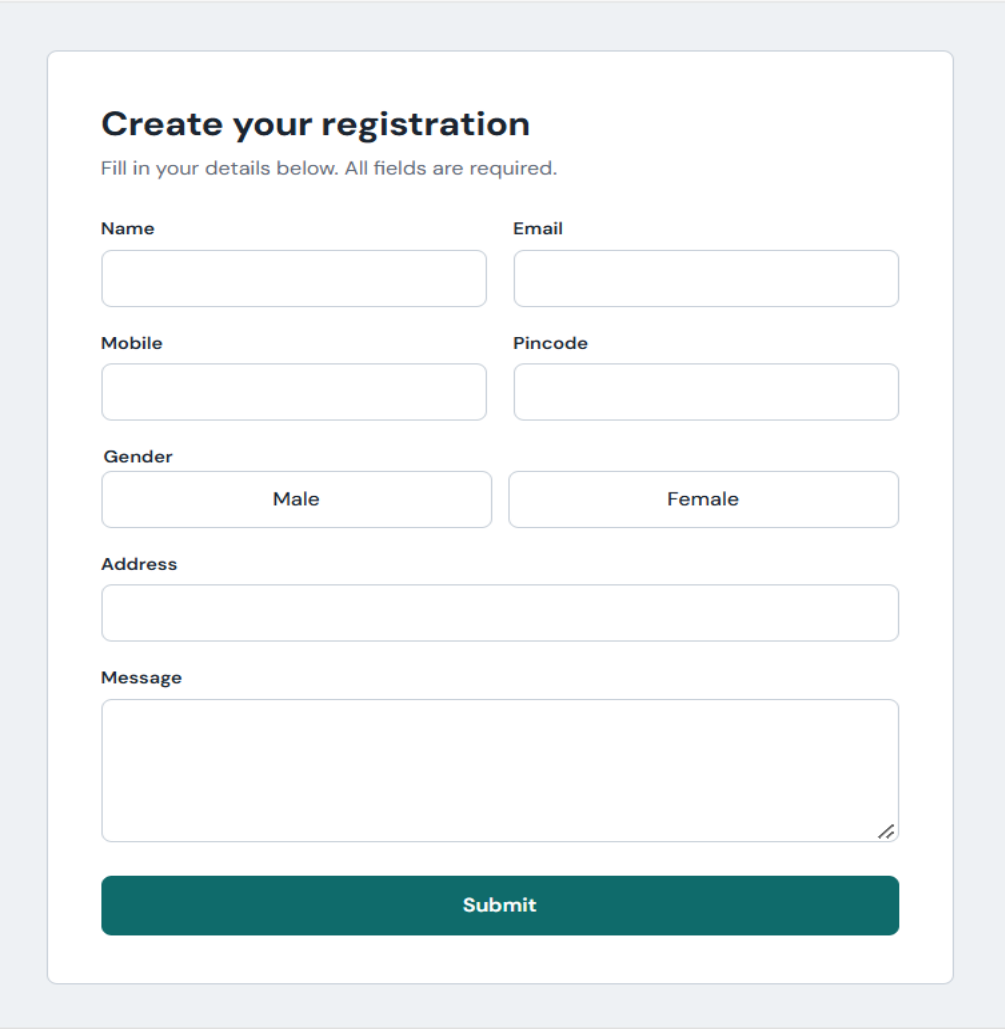
if ($storage_mode === 'csv') {
```

```
    $saved = saveToCSV('data.csv', $entry);
    $where = 'CSV';
} else {
    $saved = saveToJSON('data.json', $entry);
    $where = 'JSON';
}

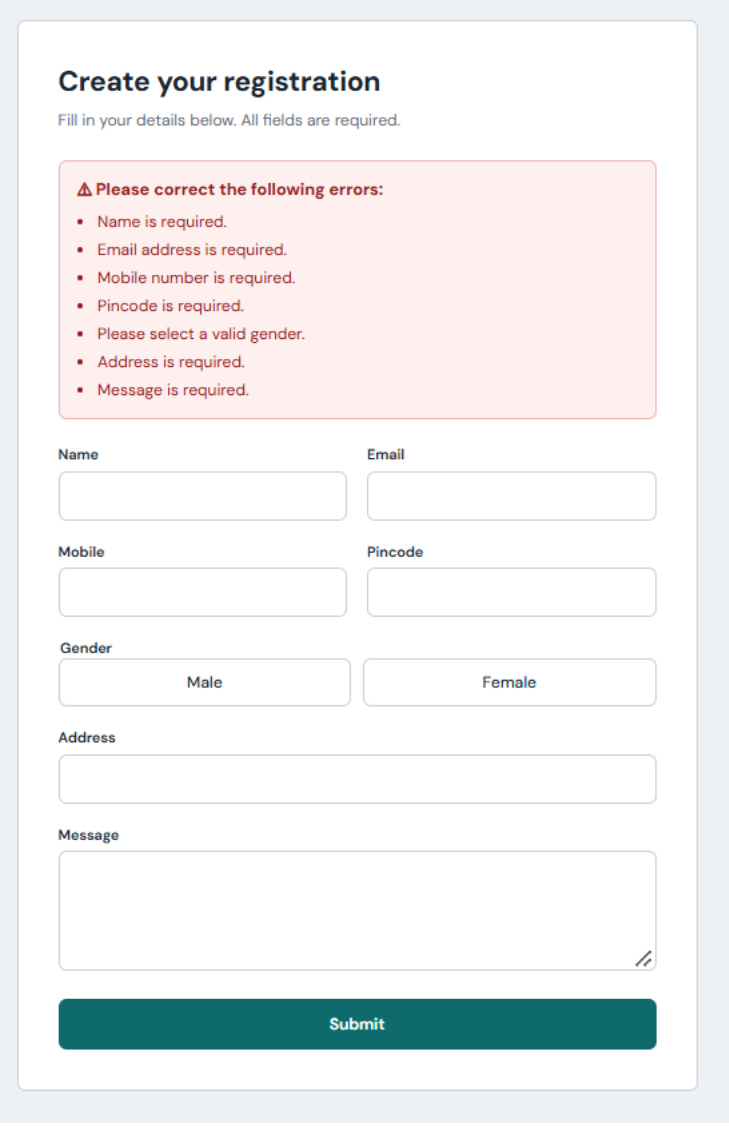
if ($saved) {
    $_SESSION['success'] = "Registration successful! Data stored in $where
file.";
} else {
    $_SESSION['errors']   = ['Unable to save registration.'];
    $_SESSION['old_data'] = $oldData;
}

header('Location: index.php');
exit();
```

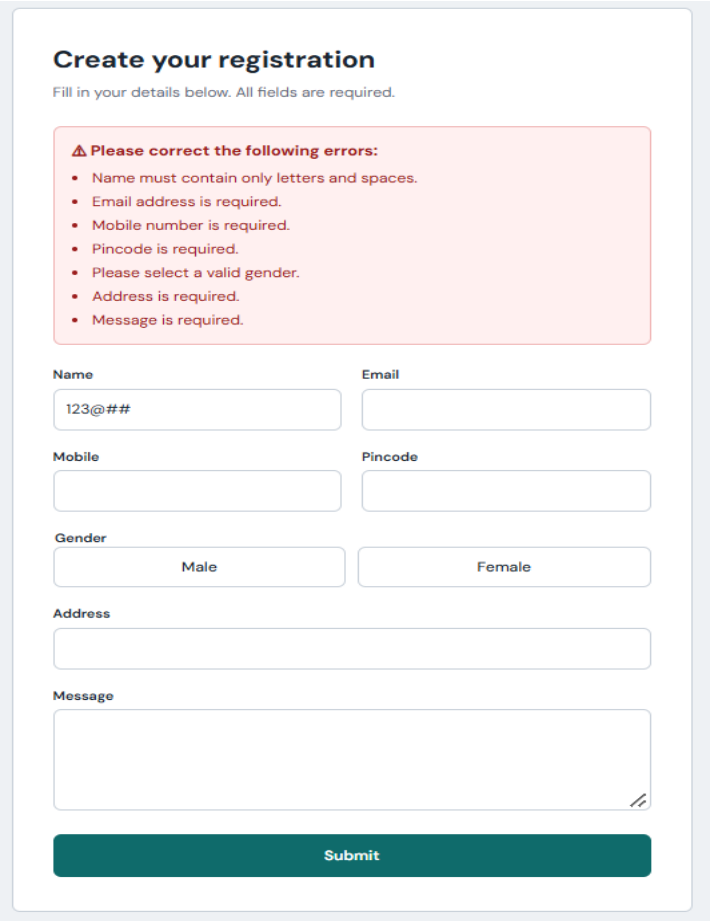
Registraition Form :



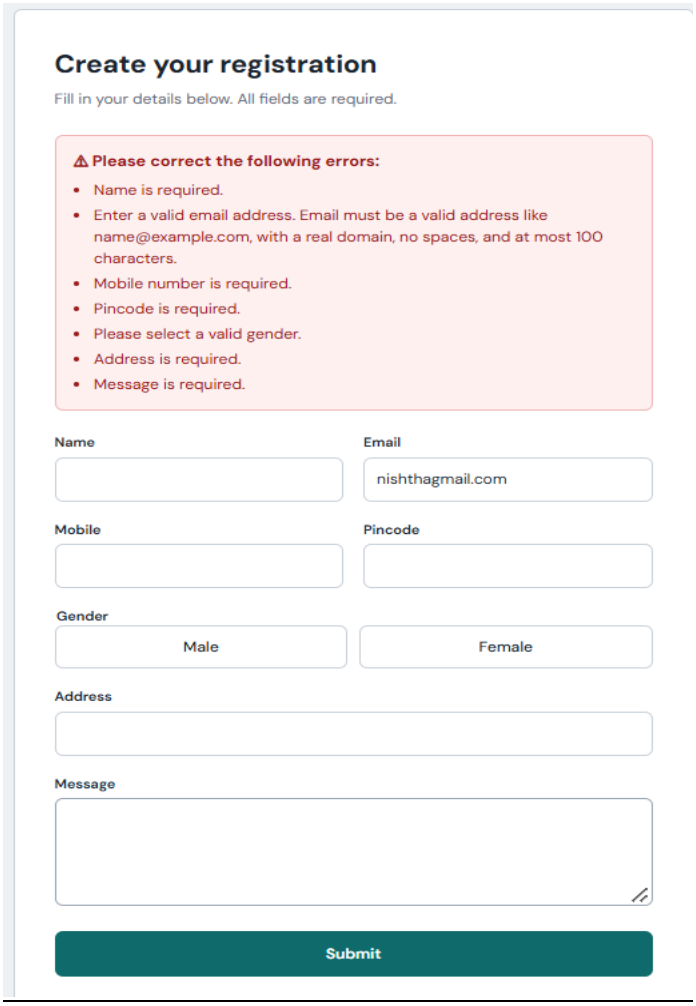
Required Field Validation :



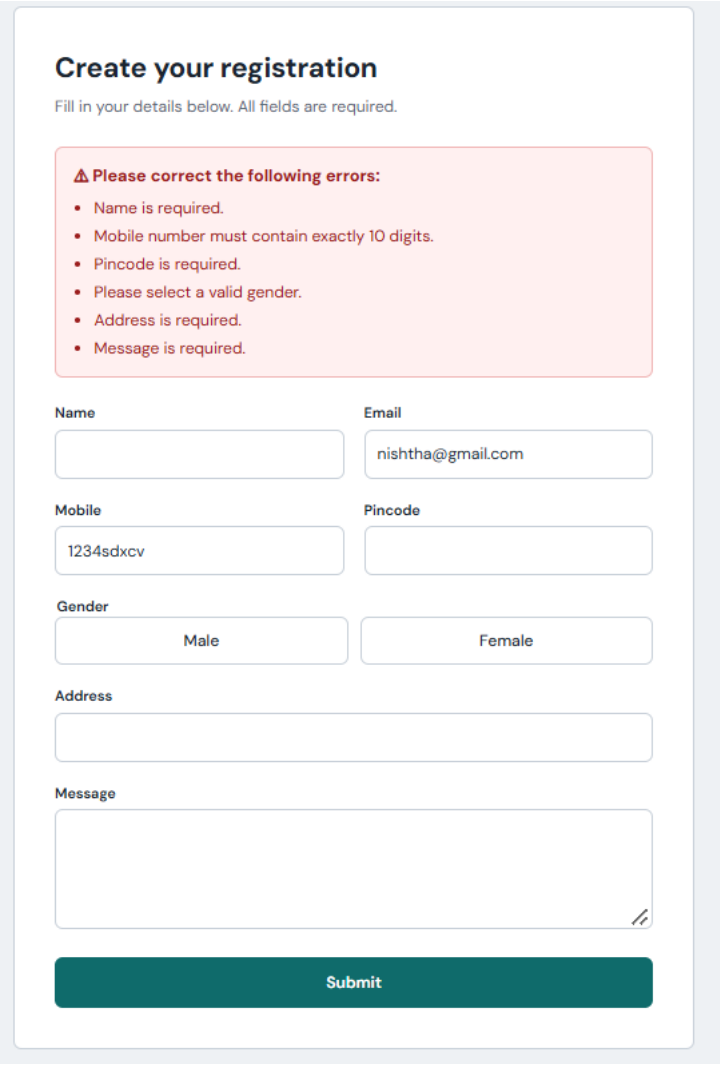
Name Format Validation :



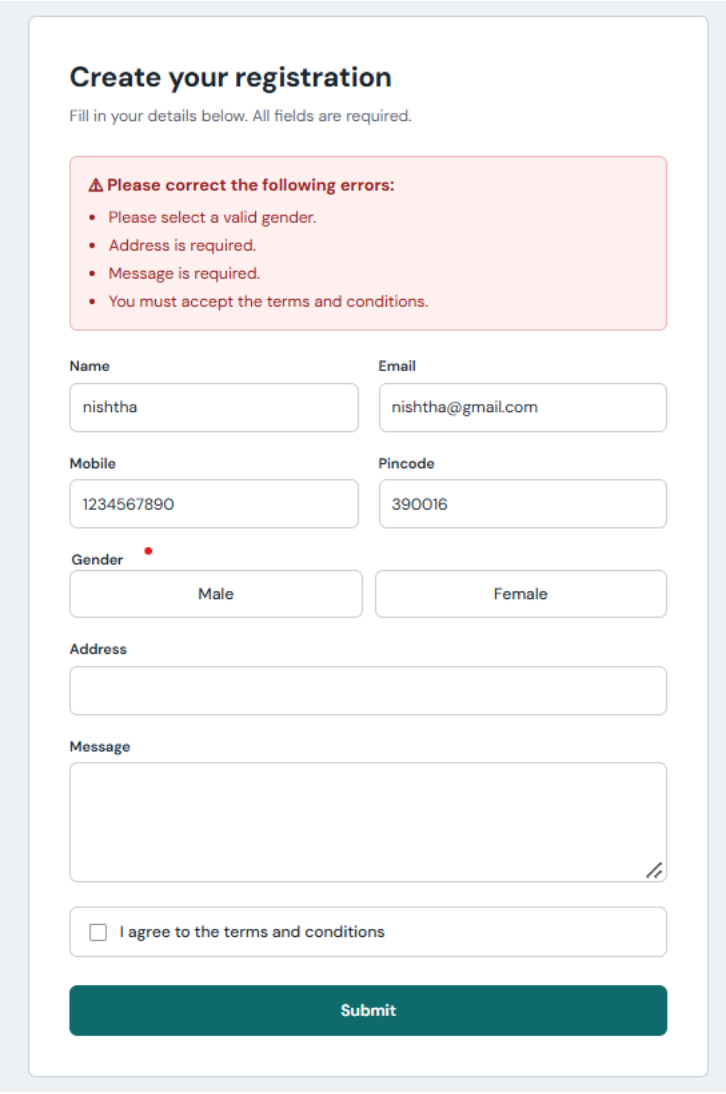
Email Format Validation :



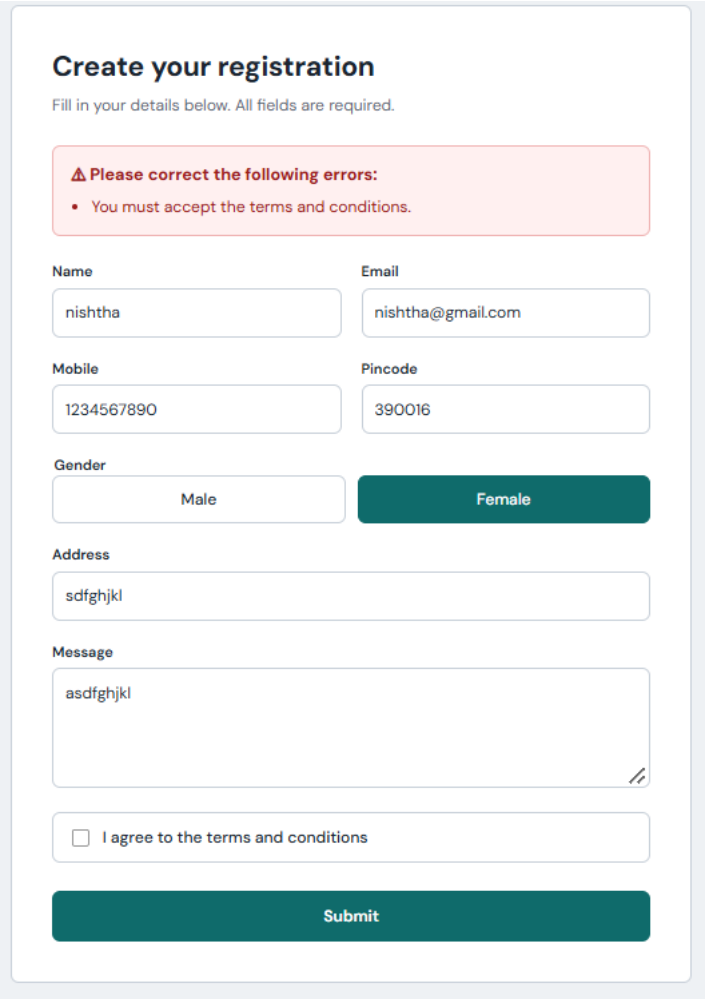
Mobile Number Format Validation :



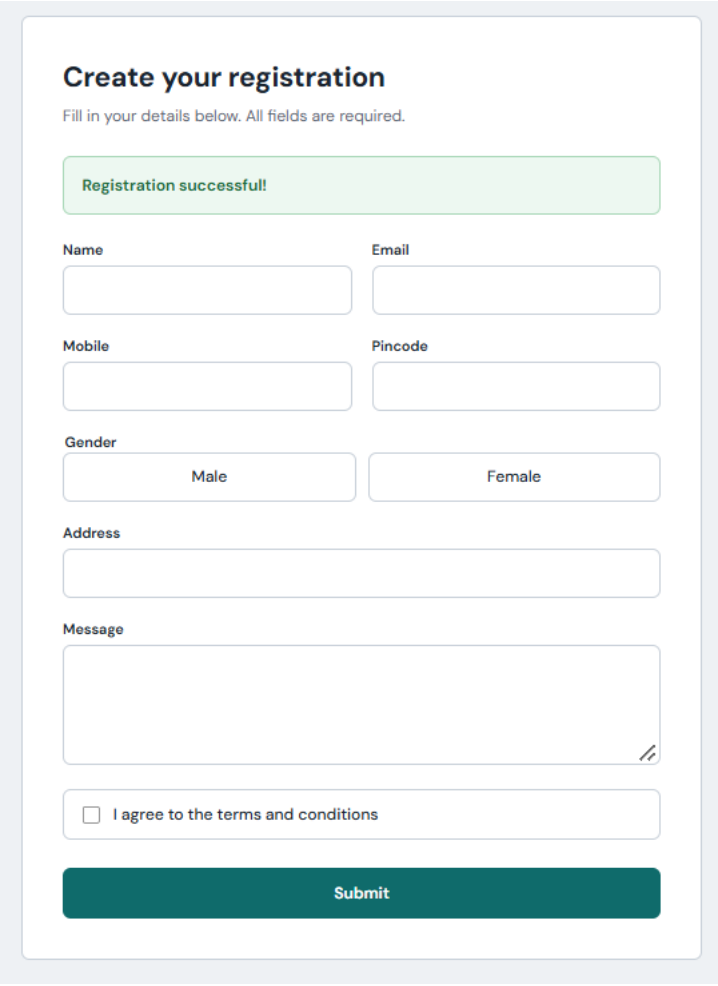
Gender Selection Validation:



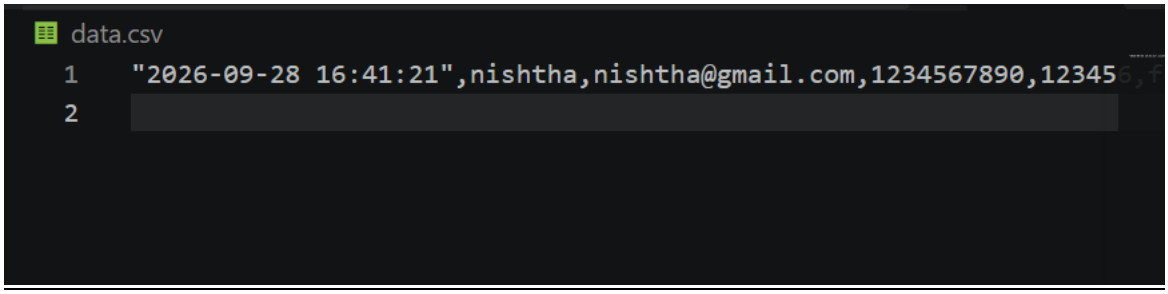
Terms & Conditions Validation :



Registration Successful :



Data Stored in CSV File successfully :



Data Stored in Json File successfully :

data.json > ...


```json
[
    {
        "timestamp": "2026-09-28 16:33:18",
        "name": "nishtha",
        "email": "nishtha@gmail.com",
        "mobile": "1234567890",
        "pincode": "390016",
        "gender": "female",
        "address": "sdfghjkl",
        "message": "asdfghjkl"
    },
    {
        "timestamp": "2026-09-28 16:34:15",
        "name": "nishtha",
        "email": "nishtha@gmail.com",
        "mobile": "1234567890",
        "pincode": "123456",
        "gender": "female",
        "address": "qwertyuio",
        "message": "qwertyuiop"
    }
]
```